

Andrei Ilin

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Education

PhD student **Université Grenoble Alpes**, Computer Architecture — Grenoble, France, 2025 – present

- Cache Coherence protocols in heterogeneous systems

Master **Université Grenoble Alpes**, Computer Science — Grenoble, France, 2023 – 2025

- MOSIG program by UGA and Ensimag
- Specialization in Parallel Computing and Distributed Systems

Bachelor **Bauman Moscow State Technical University**, Applied Mathematics and Computer Science — Moscow, Russia, 2020 – 2023

- Program with emphasis on formal methods and compiler construction

Experience

TIMA (Grenoble INP), PhD student — Grenoble, France, Dec 2025 – Dec 2028
Research on cache coherence protocols in heterogeneous systems

- Computer Architecture, GPU, Cache Coherence, Simulation, gem5

VERIMAG, Research Internship — Grenoble, France, Feb 2025 – Sept 2025
Study on timing anomalies caused by branch prediction in out of order processor

- Formal methods, Model checking, TLA+

TIMA (Grenoble INP), Research Internship — Grenoble, France, Feb 2024 – Aug 2024
Approaching automatic proofs for Chisel high level descriptions of digital circuits

- Formal proofs, Isabelle/HOL, Chisel

Huawei, Research Internship — Nizhnii Novgorod, Russia, July 2022 – Aug 2022
Optimization of video memory consumption by compressing images in Mesa3D driver

- C, Mesa3D

Projects

Formal Language Converter 2022-2023
A framework to facilitate learning formal language theory. Operations on regular expressions and finite automata and generating a step-by-step description of all transitions in LaTeX.

- Presented on OSEDUCONF-2023, a conference for educators in IT
- I contributed in leading the team of 9 people, planning the architecture and implementing some core algorithms

Path Tracing 2023

Rendering realistic images using path tracing. My coursework on the topic of Computer Graphics

- OpenGL, C++ and GLSL fragment shaders

Skills

Programming: Proficient with C++ and Git. Comfortable with C, Python, Haskell, Java, JavaScript, TypeScript, Ocaml, Golang.

Familiar with: Compilers, Cryptology, Operation Research, Computer Graphics, basics of Processor Architecture and Formal Verification, GPUs and Parallel Programming (CUDA, OpenCV, OpenMP)

Languages: English (fluent, C2), French (B2), Russian (native)

Extracurricular Activities

Game Jams: Participated in several game jams. Developed games using C++, JavaScript, TypeScript and Godot engine.